The following B.E.S.T. Standards will be covered in this unit:

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

- *Clarification 1: Problems include relationships between two chords; two secants; a secant and a tangent; and the length of the tangent from a point to a circle.*

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

- *Clarification 1: Instruction includes using the Pythagorean Theorem and completing the square.*
- *Clarification 2: Within the Geometry course, key features are limited to radius, diameter and the center.*

MA.912.GR.7.3 Graph and solve mathematical and real-word problems that are modeled with an equation of a circle. Determine and interpret key features in terms of the context.

- *Clarification 1: Key features are limited to domain, range, eccentricity, center and radius.*
- *Clarification 2: Instruction includes representing the domain and range with inequality notation, interval notation or set-builder notation.*
- *Clarification 3: Within the Geometry course, notations for domain and range are limited to inequality and set builder.*

Unit 12, Lesson 1: Solving Problems with Lengths of Intersecting Chords

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

☐ I can solve mathematical and real-world problems involving lengths formed by the intersection of two chords.

12.1.1 Warm-Up: Notice and Wonder

1. A picture of a bowling-ball pyramid is shown.

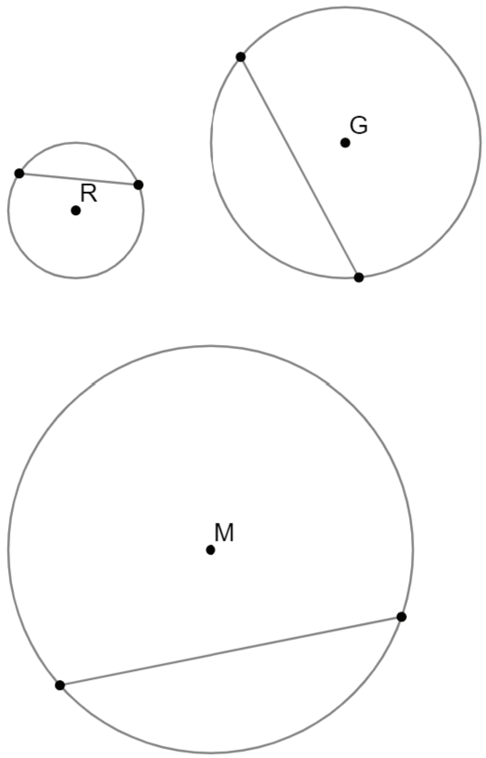
   a. What do you wonder?

   b. What do you notice?

   c. Write a question about the pyramid that a 5^{th} grader could answer.

12.1.2 Exploration: Chords

1. Circles R, G, and M are shown. Each circle has a chord. Use patty paper to copy the circles. Use the patty paper copy to construct a perpendicular bisector for each chord by folding.



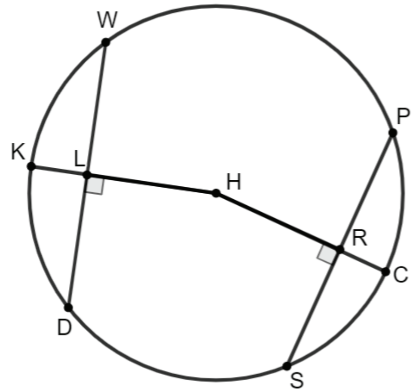
   a. Write a conjecture about the perpendicular bisector of a chord of a circle.

   b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials

   c. Review your conjecture and make any revisions, if needed.

2. Circle H is shown, where $\overline{KH}$ and $\overline{CH}$ are radii of the circle. $\overline{WD}$ and $\overline{PS}$ are chords of the circle equidistant from the center H. Therefore, $\overline{LH} \cong \overline{HR}$.



a. Using your conjecture about perpendicular bisectors, complete each congruence statement.

$\overline{DL} \cong$______

$\overline{PR} \cong$______

b. Draw a line segment from point D to point H and a line segment from point S to point H.

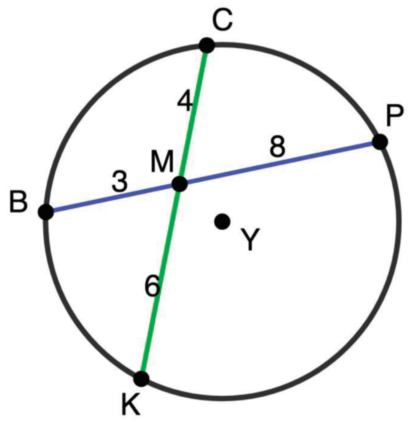
c. Justify why $\overline{DH}$ and $\overline{SH}$ are congruent.

d. Given that $\overline{LH} \cong \overline{HR}$ and $\overline{SH} \cong \overline{DH}$, what triangle congruence theorem justifies that ΔDLH and ΔSRH are congruent?

e. Since $\Delta DLH \cong \Delta SRH$, what is true about $\overline{DL}$ and $\overline{SR}$?

f. What is true about chords $\overline{DW}$ and $\overline{SP}$?

3. Circle Y is shown, where points C, P, K, and B are on circle Y. Chords $\overline{PB}$ and $\overline{KC}$ intersect at point M.



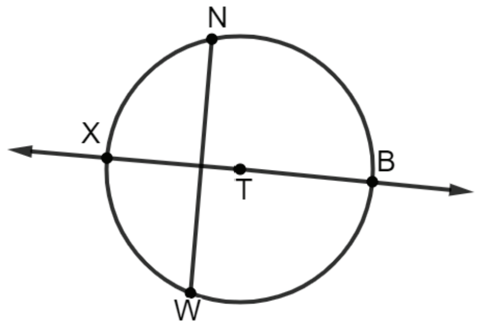
Observations of the Blue Chord Measures	Observations of the Green Chord Measures

Commonalities in the Blue and Green Chords

What do you wonder about the relationship between the blue chord and the green chord?

12.1.3 Bring It Together: Chord Relationships

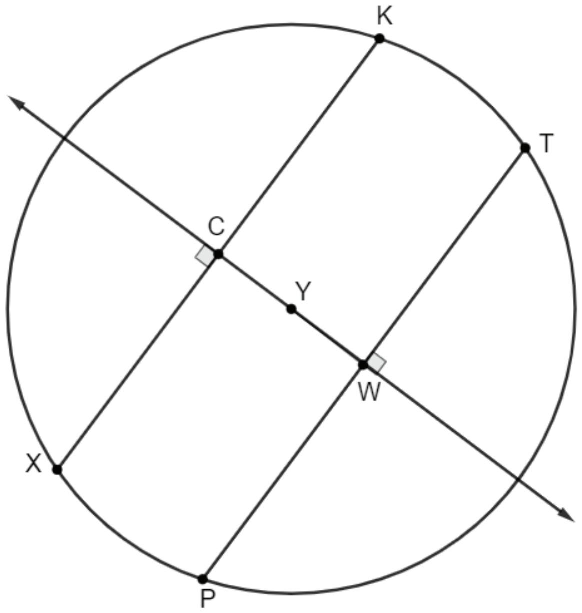
1. Circle T is shown with chord $\overline{NW}$ and secant $\overleftrightarrow{XB}$, where $\overleftrightarrow{XB}$ bisects $\overline{NW}$ and $\overline{XB}$ is a diameter of the circle.



What information is known about circle T, chord NW, and the angles formed by $\overleftrightarrow{XB}$ and $\overline{NW}$?

If a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord and its arc.

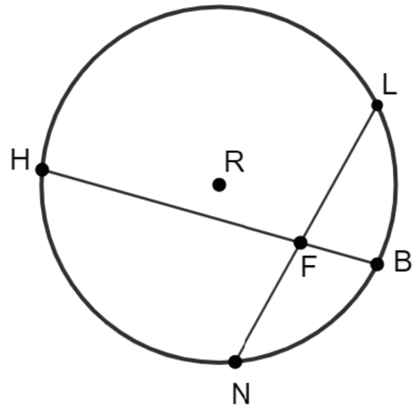
2. Circle Y is shown with chords $\overline{XK}$ and $\overline{PT}$, where $\overline{CY} \cong \overline{WY}$.



What information is known about circle Y, chords $\overline{XK}$ and $\overline{PT}$, and the angles formed by $\overleftrightarrow{CW}$?

If two chords are equidistant from the center of a circle, then the chords are congruent.

3. Circle R is shown with chords $\overline{HB}$ and $\overline{LN}$ that intersect at point F.

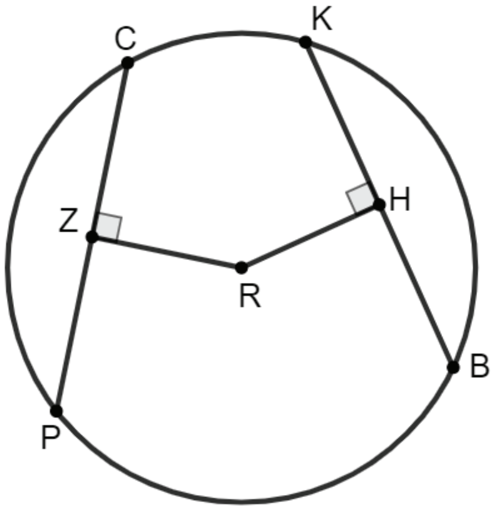


What information is known about circle R, chords $\overline{HB}$ and $\overline{LN}$, and the angles formed?

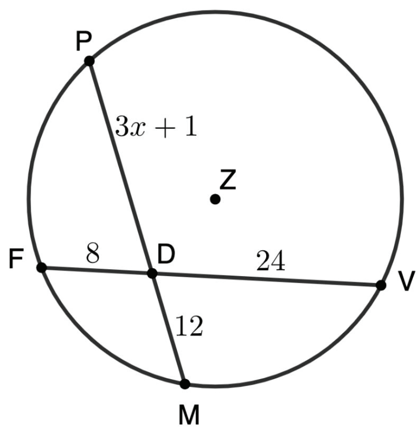
When two chords intersect each other inside a circle, the products of their segments are equal.

12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

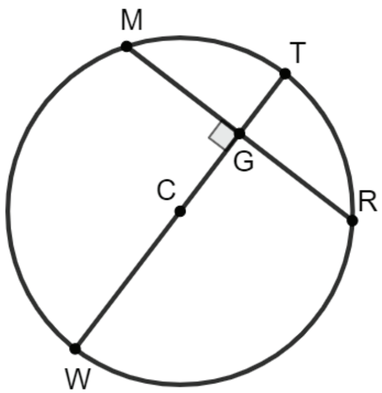
1. Circle R with points C, K, B, and P on the circle are shown with chords $\overline{CP}$ and $\overline{KB}$ equidistant from the center. Find the value of x, if $CP = 6x - 7$, $KB = 35$, $ZR = HR$, and $m\angle CZR = m\angle KHR = 90°$.



2. In circle Z, chords $\overline{PM}$ and $\overline{FV}$ intersect at point D. Points P, F, M, and V lie on circle Z. Given $FD = 8$, $DM = 12$, $VD = 24$, and $DP = 3x + 1$, solve for x.



3. Circle C is shown, where points T, R, W, and M are on the circle. The diameter, $\overline{WT}$, is a perpendicular bisector of chord $\overline{MR}$. If $MG = y + 4$ and $GR = 2y + 1$, what is the length of the chord MR?



12.1.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about solving problems involving chords.

1. I can use the proportional relationship between the measures of segments formed by intersecting chords to solve problems.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can demonstrate that chords are congruent, if the two chords are equidistant from the center of a circle.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

3. I can show that if a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Unit 12, Lesson 2: Solving Problems with Lengths of Two Secants

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

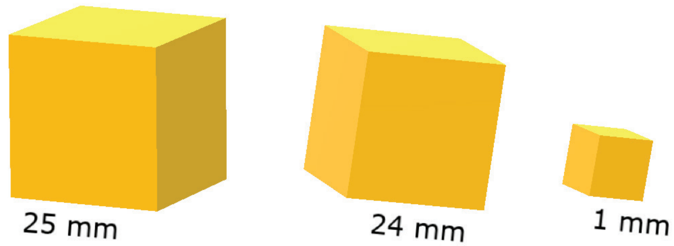
Learning Target

- [ ] I can solve mathematical and real-world problems that involve the lengths formed by the intersection of two secants.

12.2.1 Warm-Up: Would You Rather?

1. Would you rather have …

A cube of gold, 25 millimeters(mm) on each side …	OR	Two cubes of gold, one is 24 mm per side, one is 1 mm per side …

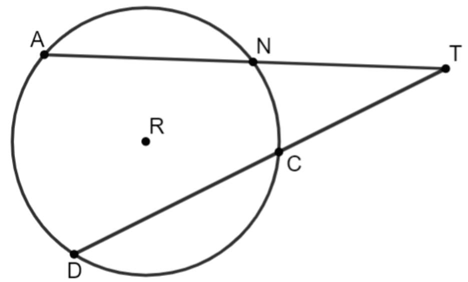


2. Explain your choice.

12.2.2 Exploration: Intersecting Secant Lengths

A ***secant*** is a line or line segment that passes through two points on a circle. In the interior of the circle, a secant line segment forms a chord.

1. Circle R is shown with points A, N, D, and C on the circle. Secants $\overline{AT}$ and $\overline{DT}$ intersect at point T outside the circle. $AN = 6$, $CD = 9$, $NT = 8$, and $CT = 7$.



   a. Complete the table.

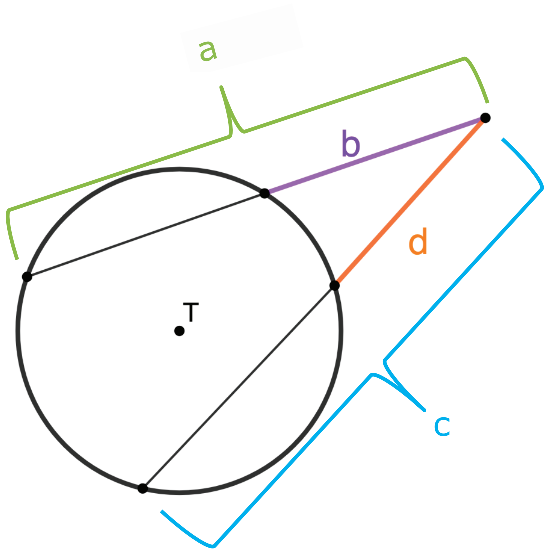
Product	Work	Answer
$AN \cdot NT$		
$AN \cdot AT$		
$AT \cdot NT$		
$DC \cdot CT$		
$DC \cdot DT$		
$DT \cdot CT$		

b. Discuss the following with your partner.

- What do you notice about your answers?

- What do you notice about the secant segments for the rows that have the same answer?

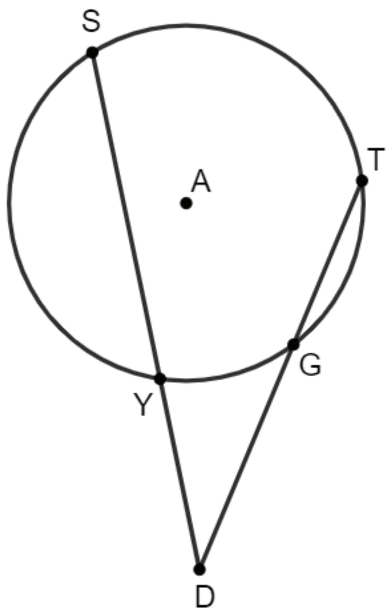
c. With your partner, use the diagram of Circle T to write a formula that can be used to find the segment lengths of intersecting secants.



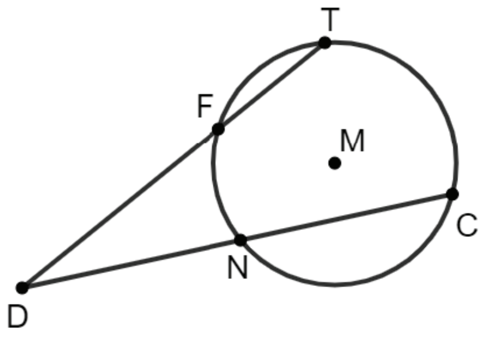
If two secants intersect outside a circle, then the product of the lengths of one secant segment and its external secant segment equals the product of the lengths of the other secant segment and its external secant segment.

12.2.3 Guided Practice: Solving Problems Involving Intersecting Secant Lengths

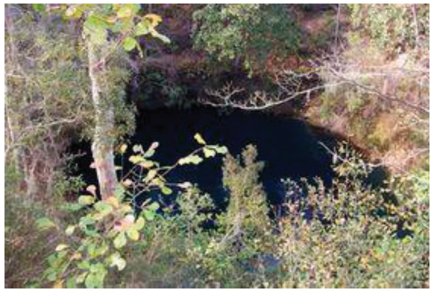
1. Circle A with secants $\overline{SD}$ and $\overline{TD}$ is shown. $\overline{SD}$ and $\overline{TD}$ intersect at point D outside the circle so that $SY = 5$, $DY = 3$, $TG = x$, and $DG = 4$. Solve for x.

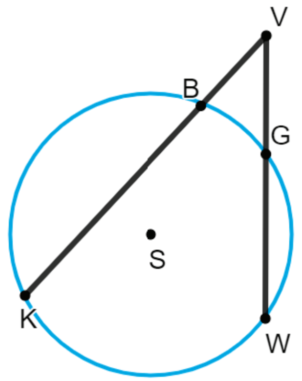


2. Circle M is shown, where points T, F, N, and C are on circle M. Secants $\overline{TD}$ and $\overline{CD}$ intersect at point D outside the circle such that $TF = 7$, $FD = 9$, $CN = 5x$, and $ND = 4x$. Determine the value of x.



3. Leon Sinks Geologic Area is located in Apalachicola National Forest. The area has 18 sinkholes, many of which create a cave system. A park ranger mapped a circular sinkhole. His diagram is shown, where circle S has secants $\overline{VK}$ and $\overline{VW}$ such that points B and G are on circle S and point V is outside the circle. $KB = (4x + 4.25)$ meters (m), $VB = 12.75$ m, $GW = (3x + 5.5)$ m, and $GV = 14$ m. Find the distance, in meters, from point K to point B.

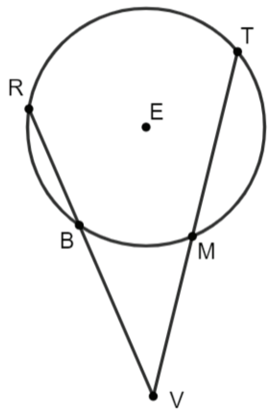




12.2.4 Your Turn!

1. Circle E has points R, F, G, and M on the circle. Secants $\overline{RZ}$ and $\overline{MZ}$ intersect at point Z outside the circle. Point G is on $\overline{MZ}$, and point F is on $\overline{RZ}$. $MG = 10$, $GZ = 8$, $RF = 2x - 17$, and $FZ = 9$. Solve for x.

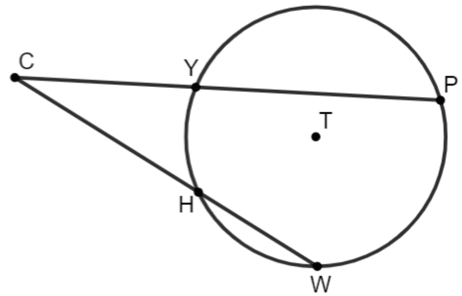
2. Tammie was calculating the distance between her favorite places to visit in Epcot. Circle E represents the World Showcase where R, T, M, and B are on the circle and point V is outside the circle at Spaceship Earth. Tammie knows that $RB = 975$ feet (ft), $BV = 1425$ ft, and $VM = 1250$ ft. What is the length of $\overline{MT}$, in feet?



12.2.5 Wrap-Up: Error Analysis

1. Donald was finding the value of x in circle T, where Y, P, W, and H are points on the circle. He knows that $\overline{CP}$ and $\overline{CW}$ are secants of the circle, where $CY = 4$, $YP = x - 3$, $CH = 5$, and $HW = x - 6$. His work is shown.

Donald's Work
$4(x-3) = 5(x-6)$
$4x - 12 = 5x - 30$
$x = 18$



Write a note to Donald explaining the mistake he made when solving for x.

Unit 12, Lesson 3: Solving Problems with Secant and Tangent Segments

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can solve mathematical and real-world problems that involve lengths formed by the intersection of a secant and a tangent segment.

12.3.1 Warm-Up: Notice and Wonder

1. The Seminole Native American tribe is known for their novel and colorful dress consisting of patchwork. The decorative technique of patchwork was developed by Seminole women before 1920. Patchwork was rapidly adopted to further embellish the already colorful clothing. As time went on, the designs became more intricate as the seamstresses became more skilled.

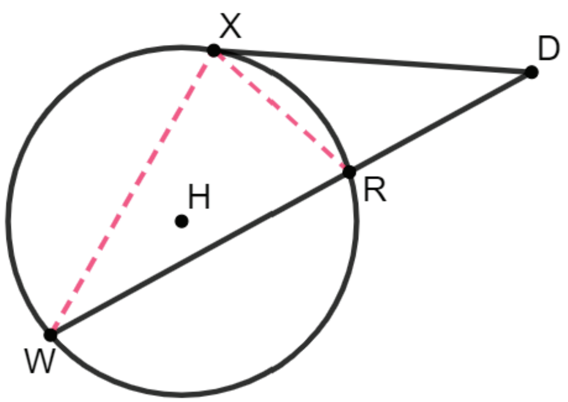
   a. What do you notice about the patchwork?

   b. What do you wonder?

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

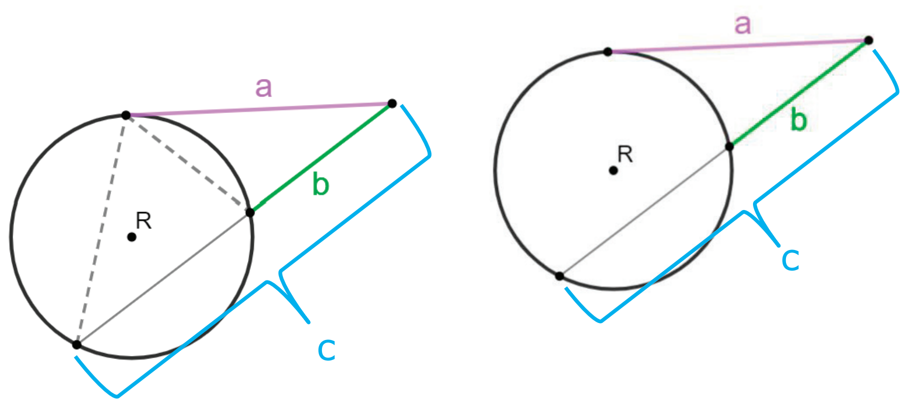
Work with your partner to complete the following.

1. Circle H has points X, R, and W on the circle. Secant $\overline{WD}$ and tangent $\overline{XD}$ intersect at point D outside the circle. Given that $\Delta DWX \sim \Delta DXR$, $XD = 15$, and $RD = 9$.



   a. Use the similiar triangles to write a proportional relationship that can be used to determine the length of $\overline{RW}$.

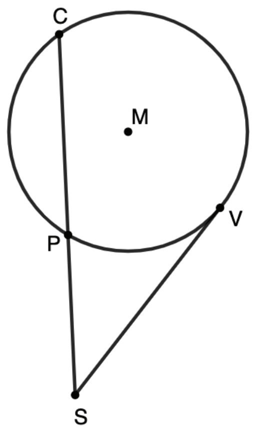
   b. Use the diagrams of circle R to write a formula that can be used to find the segment lengths of the intersecting secant and tangent lines.



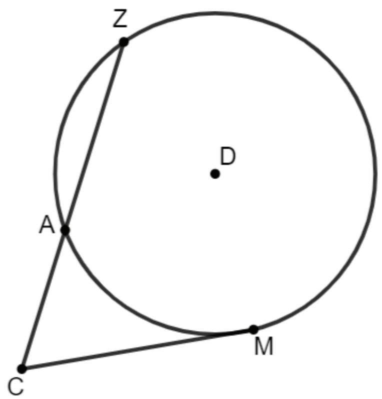
If a secant and tangent segment are drawn to a circle from the same external point, then the square of the measure of the tangent segment is equal to the product of the measures of the secant segment and its external secant segment.

12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

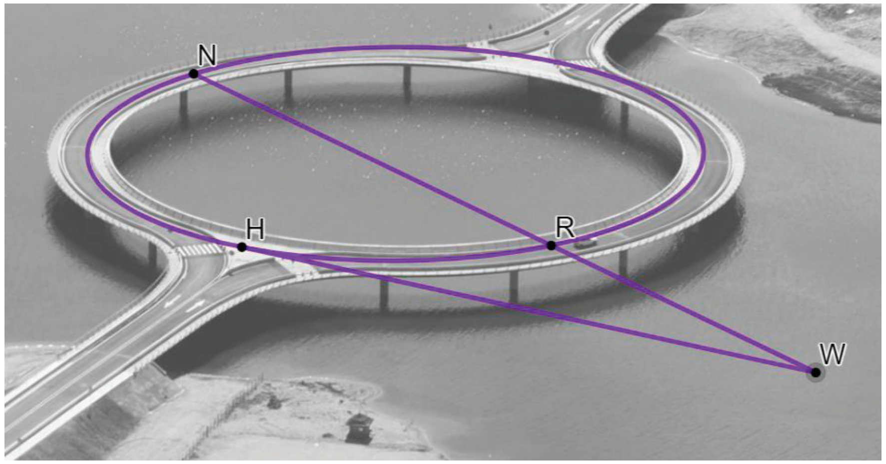
1. Circle M has points C, V, and P on the circle. Secant $\overline{CS}$ and tangent $\overline{VS}$ intersect at point S outside the circle. $CP = 4$ and $SP = 3$. What is the length of $\overline{SV}$? Round to the nearest thousandth.



2. Circle D is shown with points Z, A, and M on the circle. Secant $\overline{CZ}$ and tangent $\overline{CM}$ intersect at point C outside the circle. $AZ = 16$, $AC = x - 3$, and $MC = x + 3$. What is the length of $\overline{CM}$?

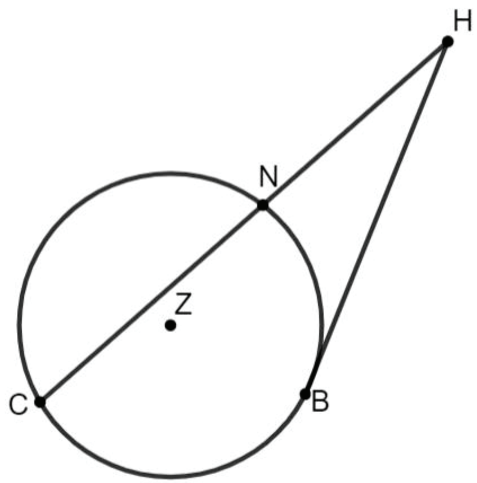


3. A drone flew over the Laguna Garzon Bridge in Uruguay and took pictures at three points on the bridge: N, R, and H. Tangent $\overline{HW}$ and secant $\overline{NW}$ intersect at point W off the bridge. If $HW = 280$ feet (ft) and $RW = 160$ ft, what is the length of $\overline{NW}$?

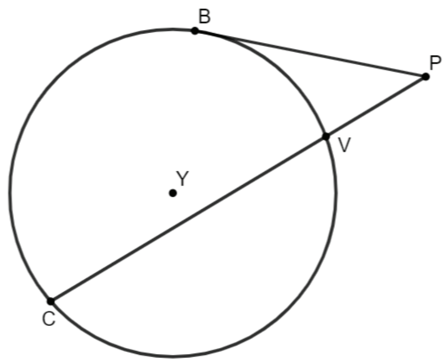


12.3.4 Your Turn!

1. Circle Z is shown with points C, N, and B on the circle. Secant $\overline{CH}$ and tangent $\overline{BH}$ intersect at point H outside the circle. $CN = 7x - 6$, $NH = 12$, and $HB = 18$. Find the value of x.



2. Isabella and Gregory were working on finding the length of $\overline{BP}$. Circle Y is given, where B, V, and C are points on the circle. Secant $\overline{CP}$ and tangent $\overline{BP}$ intersect at point P outside the circle. $CV = 12$ and $VP = 4$.



Isabella's and Gregory's work to find the length of $\overline{BP}$ is shown.

Isabella's Work	Gregory's Work
$x^2 = 4(4+12)$ $x^2 = 4(16)$ $x^2 = 64$ $x = 8$	$x^2 = 12(4+12)$ $x^2 = 12(16)$ $x^2 = 192$ $x = 13.9$

a. Whose work do you agree with?

b. Write a note to the person whose work is incorrect explaining their mistake.

12.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 12, Lesson 4: Solving Problems with Lengths of Two Tangents

Benchmark

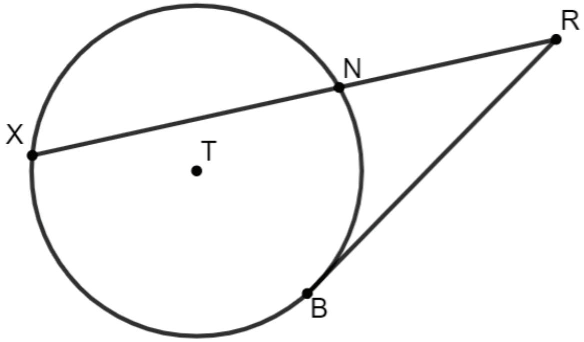
MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- [ ] I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.

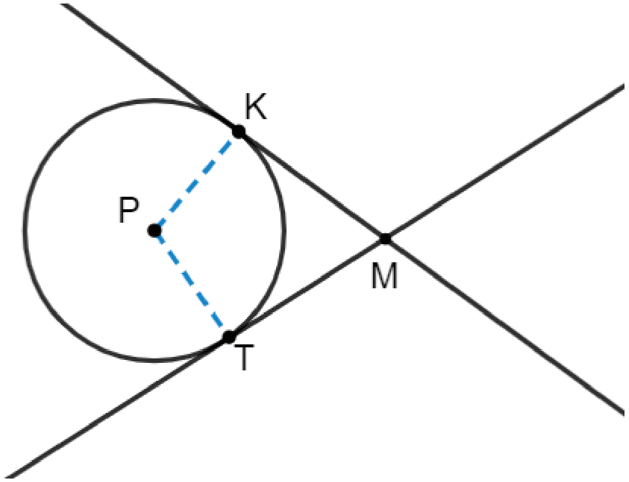
12.4.1 Warm-Up: Bell Work

1. Circle T has points X, N, and B on the circle. Secant $\overline{XR}$ and tangent $\overline{BR}$ intersect at point R outside the circle. Given that $XN = 16$ and $NR = 9$, find the length of $\overline{BR}$.



12.4.2 Exploration: Intersecting Tangent Lines

1. Circle P is shown with tangents $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ and radii $\overline{PK}$ and $\overline{PT}$.



   a. Draw a dotted line from point P to point M.

   b. Discuss with your partner the relationships that can be determined given that $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ are tangents, and $\overline{PK}$ and $\overline{PT}$ are radii.

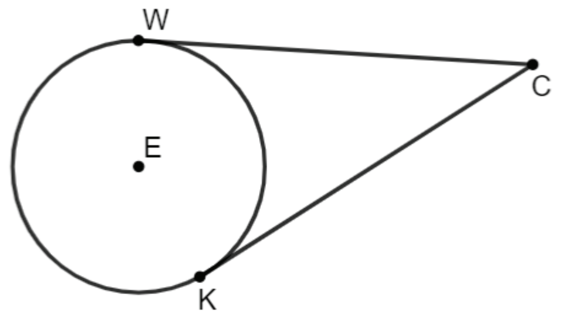
c. Discuss with your partner why HL congruence can be used to show that ΔMKP and ΔMTP are congruent.

d. Since $\Delta MKP \cong \Delta MTP$, what is true about $\overline{MK}$ and $\overline{MT}$?

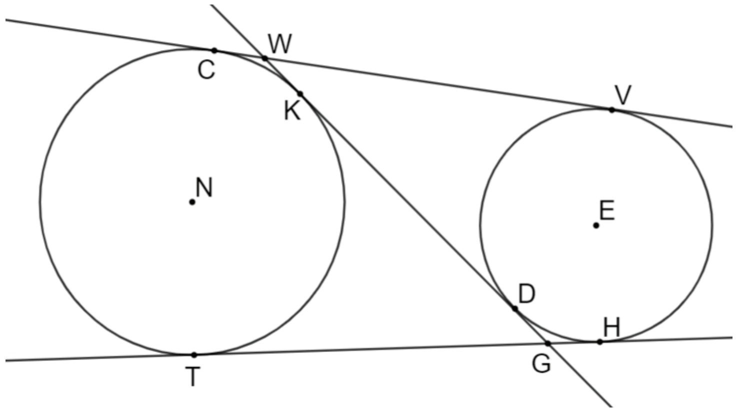
Tangent segments to a circle from the same point are congruent.

12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

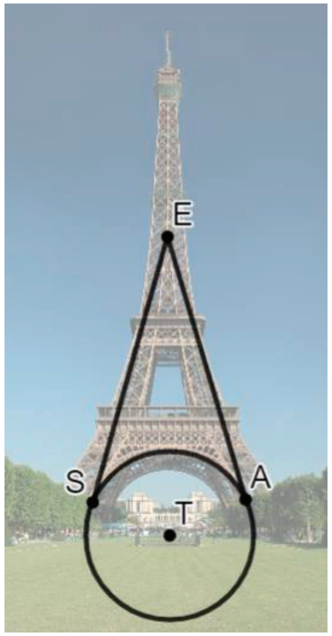
1. Circle E has points W and K on the circle. Tangents $\overline{WC}$ and $\overline{KC}$ intersect at point C outside the circle. If $WC = x + 15$ and $KC = 2x - 5$, find the length of $\overline{KC}$.



2. Circle N has points C, K, and T on the circle. Circle E has points V, H, and D on the circle. Tangents $\overline{TG}$ and $\overline{KG}$ intersect outside the circle at point G. Tangents $\overline{VW}$ and $\overline{WD}$ intersect outside the circle at point W. If $TG = (7x - 36)$, $KG = (5x - 14)$, $VW = (8y + 13)$, and $WD = KG$, what is the value of y?

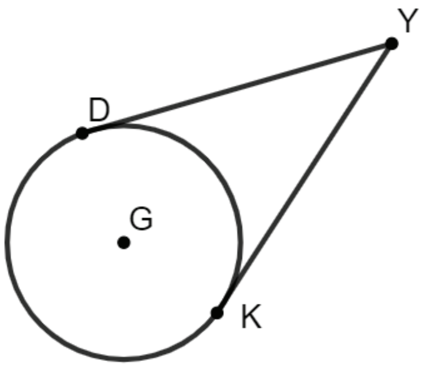


3. The Eiffel Tower in Paris, France stands at 324 meters tall. The tower has four semicircular arches at its base. Circle T represents one of the four arches at the base of the tower, where points S and A lie on the circle. If $SE = 17x + 27$ meters (m) and $EA = 20x - 15$ m, what is the length of $\overline{EA}$ in meters?

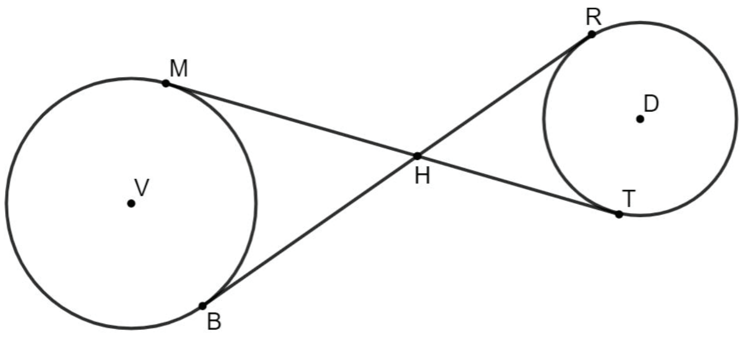


12.4.4 Your Turn!

1. Circle G has points D and K on the circle. Tangents $\overline{DY}$ and $\overline{KY}$ intersect at point Y outside the circle. If $DY = 4x - 19$ and $KY = 15$, find the value of x.



2. Points M and B are on the circle V and points R and T are on the circle D. $\overline{MT}$ and $\overline{BR}$ intersect at point H. $\overline{MT}$ is tangent to circle V at point M and tangent to circle D at point T. $\overline{BR}$ is tangent to circle V at point B and tangent to circle D at point R. If $MH = 4x - 8$, $BH = 2x + 1$, $TH = 2x - 1$, and $RH = 5y - 7$, what is the value of y?



12.4.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Unit 12, Lesson 5: Pythagorean Theorem and Tangent Lines

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- [ ] I can use the Pythagorean Theorem to solve mathematical and real-world problems involving the length of a tangent segment to a circle.

12.5.1 Warm-Up: Ice Problem

1. An igloo, also known as a snow house or snow hut, is a type of shelter built out of snow.

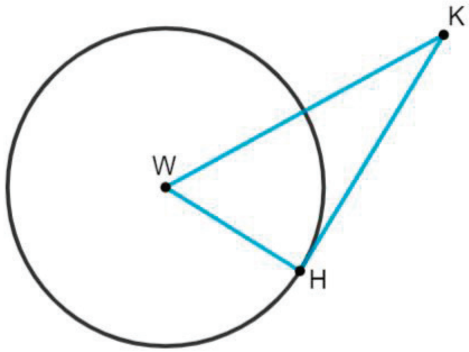
   a. What is the first question that comes to your mind?

   b. Write a question that would require someone to use geometry to solve.

12.5.2 Exploration: Pythagorean Theorem and Tangent Lines

Work with your partner to complete the following.

1. Circle W has point H on the circle. $\overline{WK}$ and tangent $\overline{KH}$ intersect at point K outside the circle.



   a. Discuss with your partner what can be concluded from the given information. Justify your conclusions. Give at least two conclusions.

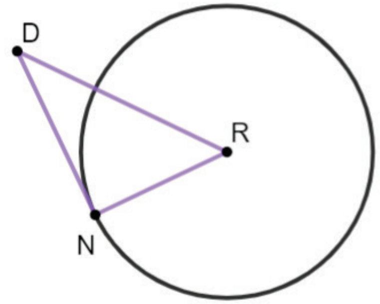
b. Ask a classmate for two conclusions they drew. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	**My Summary of Their Reason**	**Initials**

c. Discuss with your partner what information is needed to determine the length of $\overline{WK}$.

d. Explain why any triangle drawn, where the radius and a tangent are the legs, will result in a right triangle.

2. Circle R is shown with point D outside the circle, where point N is on the circle. $DN = 10$, $NR = 8$, and $DR = 4\sqrt{10}$.



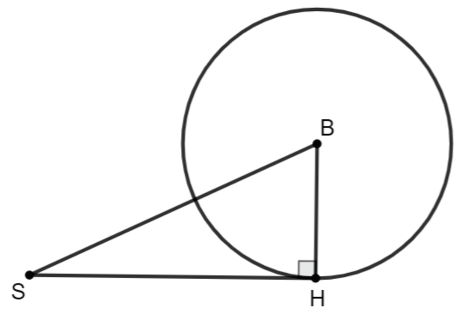
a. Use the Pythagorean Theorem to determine if $\overline{DN}$ is a tangent to circle R.

b. Complete the statements.

ΔDRN ○ is ○ is not a right triangle. Therefore $\overline{DN}$ ○ is ○ is not a tangent line to circle R.

12.5.3 Guided Practice: Pythagorean Theorem and Tangent Lines

1. Circle B has point H on the circle and point S outside the circle. $m\angle SHB = 90°$, $SB = 13$, and $BH = 5$.

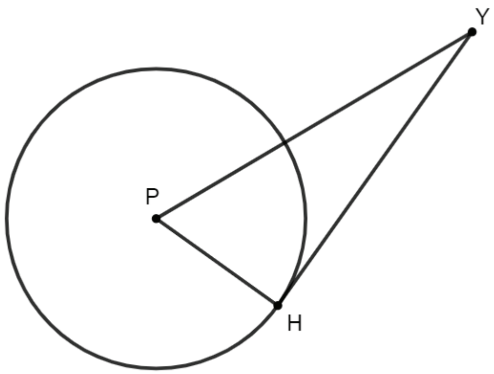


   a. What is the length of $\overline{SH}$?

   b. What is the area of ΔSBH?

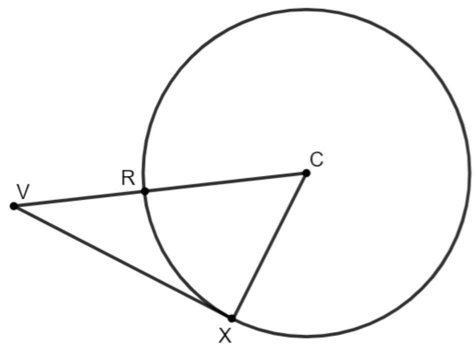
2. Circle P has point H on the circle and point Y outside the circle, where $\overline{PY}$ and tangent $\overline{HY}$ intersect. $PH = 10$ and $HY = 17$.

   What is the length of $\overline{PY}$?



3. Circle C has points R and X on the circle and point V outside the circle, where $\overline{VC}$ and tangent $\overline{VX}$ intersect at point V. $VX = 24$ and $VR = 16$.

   a. What is the length of $\overline{CX}$?

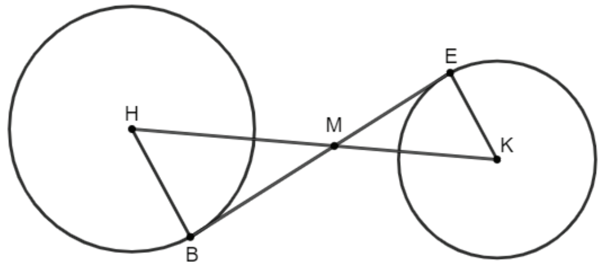


   b. What is the length of $\overline{VC}$?

   c. What is the perimeter of ΔVCX?

12.5.4 Your Turn!

1. Circle H has point B on the circle and circle K has point E on the circle. $\overline{HK}$ and tangent $\overline{EB}$ intersect outside the circles at point M. $m\angle HBM = m\angle MEK = 90°$, $HB = 6$, $BM = 8$, and $MK = 8$.



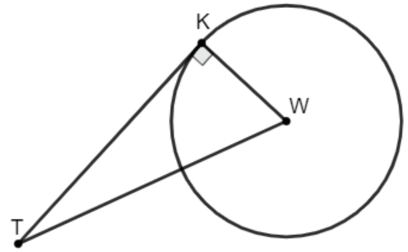
   a. Explain why $\Delta HMB \sim \Delta KME$.

   b. What is the length of $\overline{HK}$?

   c. What is the perimeter of ΔKME?

   d. What is the area of ΔKME?

2. Grace and Zackary are finding the length of tangent $\overline{TK}$ of circle W, where K is a point on the circle. $\overline{TW}$ and tangent $\overline{TK}$ intersect at point T outside the circle. $TW = 18$ and $KW = 7$.



Grace's and Zackary's work is shown.

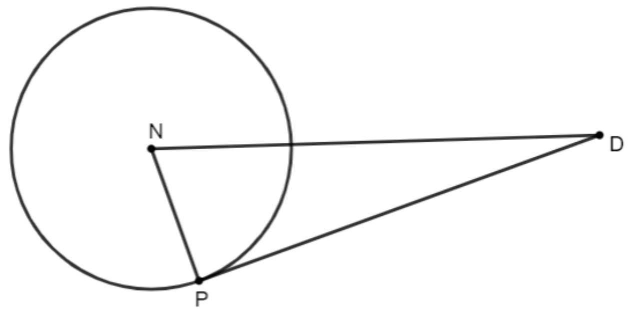
Grace's Work	Zackary's Work
$7^2 + x^2 = 18^2$ $49 + x^2 = 324$ $x^2 = 275$ $x = 5\sqrt{11}$	$7^2 + 18^2 = x^2$ $49 + 324 = x^2$ $x^2 = 373$ $x = \sqrt{373}$

Write a note to the person whose work is incorrect explaining their mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

12.5.5 Wrap-Up: Ticket Out the Door

1. Circle N has point P on the circle and point D outside the circle at the intersection of $\overline{DN}$ and tangent $\overline{DP}$. $NP = 2$ and $DP = 6$.



What is the length of $\overline{ND}$?

Unit 12, Lesson 6: Lengths of Secants, Tangents, Segments, and Chords in Circles

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can solve mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

12.6.1 Warm-Up: Critical Thinking

1. Explain how math could be considered a language used for communication.

2. Suppose you were taking Geometry in another country. How do you think the problems would differ?

12.6.2 Activity: Line Segments of Circles

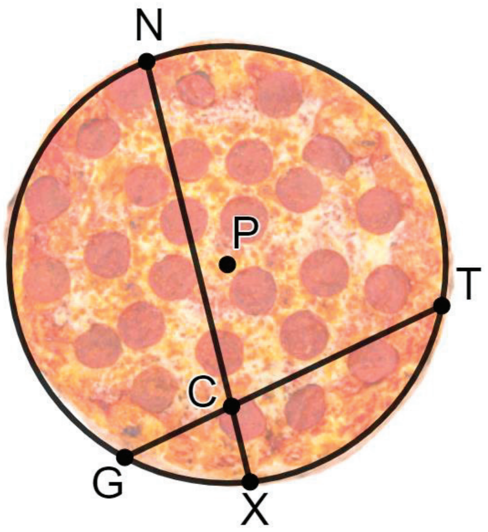
Match each image, type of segment, and answer to the description of circles. Record your answers in the table.

Circle	Work	Cards' Letter
Circle K has points G, N, D, and E on the circle, where $GM = 2x + 2$, $MD = 14$, $EM = 12$, and $MN = 2x + 5$		Image: ____ Segment: ____ Answer: ____
Circle H has points C, T, and B on the circle, where $BR = 15$, $RT = 9$, and $CT = x$		Image: ____ Segment: ____ Answer: ____
Circle R has points T, S, V, and W on the circle, where $TS = x - 8$, $SP = 4$, $VP = 3$, and $VW = x - 5$		Image: ____ Segment: ____ Answer: ____

Circle	Work	Cards' Letter
Circle F has points E and B on the circle, where $EY = 7x - 3$ and $BY = 5x + 1$		Image: ____ Segment: ____ Answer: ____
Circle V has point B on the circle, where $VB = 4$, $BA = 8$ and $VA = x$		Image: ____ Segment: ____ Answer: ____
Circle A has points G, R, H, and S on the circle, where $MA = WA$, $GR = 35$, and $SH = 6x - 7$		Image: ____ Segment: ____ Answer: ____

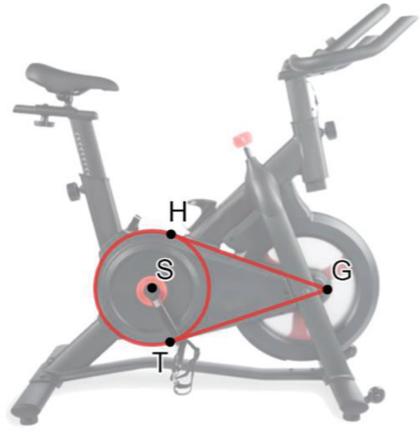
12.6.3 Try It: Solving for Length

1. When Doug's pizza was delivered, he realized the pizza was not cut evenly. The pizza, circle P, is cut by chords $\overline{NX}$ and $\overline{GT}$.



Given that $NC = (5x - 6)$ inches (in.), $CX = 2$ in., $GC = 3$ in., and $CT = 2x$ in., what is the length of $\overline{NX}$?

2. A belt guard is used on stationary bikes to cover some of the inner mechanics. Circle S is shown, where points H and T are on the circle. Tangents $\overline{HG}$ and $\overline{TG}$ intersect outside the circle at point G.



If $HG = 15$ and $TG = 4y - 9$, what is the value of y?

12.6.4 Wrap-Up: Cool Down

1. Write **one question** you still have about solving mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Unit 12, Lesson 7: Deriving the Equation of a Circle

Benchmark

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

Learning Targets

- ☐ I can use the Pythagorean Theorem to derive the equation of a circle.
- ☐ I can determine key features of a circle given the graph or equation.
- ☐ I can write the equation of a circle using key features.

12.7.1 Warm-Up: Roundabouts

Roundabouts promote a continuous flow of traffic because drivers do not have to wait for a green light at a roundabout to get through the intersection. A diagram of a roundabout is shown.

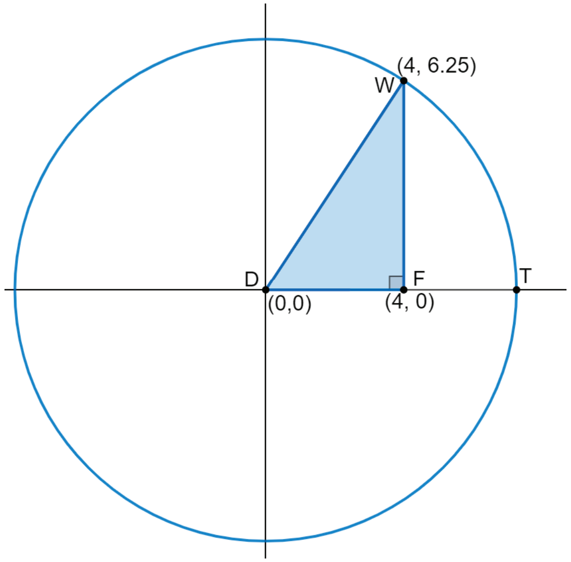
1. What do you wonder about the diagram?

2. How many different paths could the red car take?

3. List at least two things you have studied in Geometry so far that may be helpful to know when constructing a roundabout.

12.7.2 Exploration: Deriving the Equation of a Circle

1. The center of circle D is located at $(0,0)$. Point W lies on circle D at the point $(4, 6.25)$. Right triangle DFW is drawn inside the circle, such that $\overline{DW}$ is a circle's radius and point F, $(4,0)$, lies on radius $\overline{DT}$.



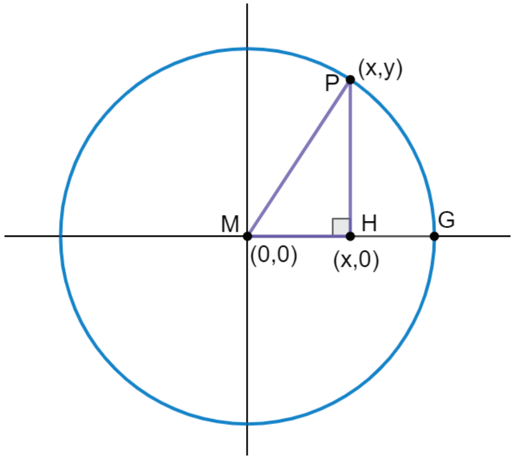
a. Find the length of each side.

Line Segment	Distance Between Endpoints
$\overline{DF}$	
$\overline{FW}$	

b. Use the values you found for the lengths of $\overline{DF}$ and $\overline{FW}$ to complete the equation that can be used to determine the length of $\overline{DW}$ by applying the Pythagorean Theorem.

$$(DW)^2 = (\qquad)^2 + (\qquad)^2$$

2. The center of circle M is located at $(0,0)$. Point P lies on circle M at the point (x, y). Right triangle MPH is drawn inside the circle, such that $\overline{MP}$ is a circle's radius and point H, $(x, 0)$, lies on radius $\overline{MG}$.



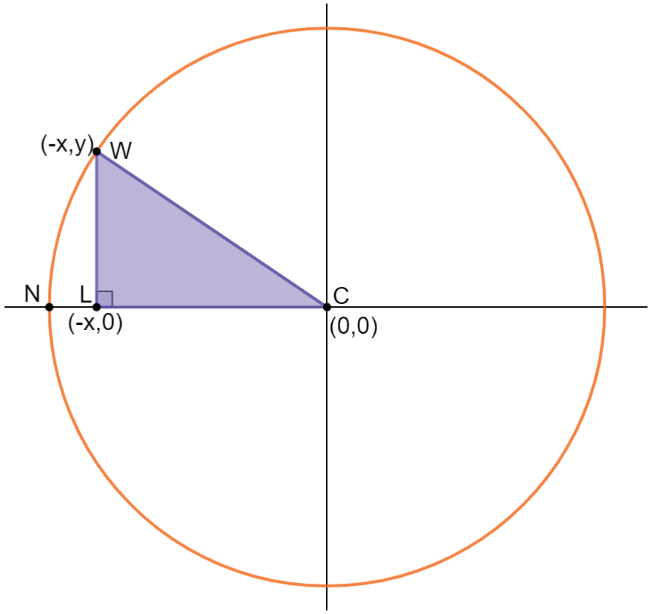
a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{MH}$	
$\overline{HP}$	

b. Use the expressions you found for the lengths of $\overline{MH}$ and $\overline{HP}$ to complete the equation that can be used to determine the length of $\overline{MP}$ by applying the Pythagorean Theorem.

$(MP)^2 = (\qquad)^2 + (\qquad)^2$

3. The center of circle C is located at $(0,0)$. Point W lies on circle C at the point $(-x, y)$. Right triangle WLC is drawn inside the circle, such that $\overline{WC}$ is a circle's radius and point L, $(-x, 0)$, lies on radius $\overline{NC}$.



a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{LC}$	
$\overline{LW}$	

b. Use the expressions you found for the lengths of $\overline{LC}$ and $\overline{LW}$ to complete the equation that can be used to determine the length of $\overline{WC}$ by applying the Pythagorean Theorem.

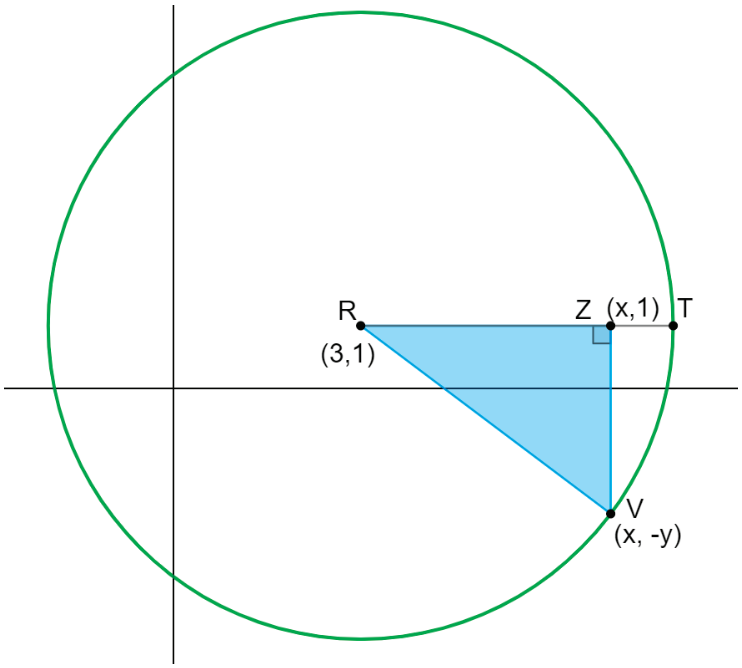
$(WC)^2 = (\qquad)^2 + (\qquad)^2$

c. Complete the table with the equation you found that can be used to determine the length of the radius of each circle.

Line Segment	Distance
$\overline{MP}$ radius of circle M	$(MP)^2 =$ or $(r)^2 =$
$\overline{WC}$ radius of circle C	$(WC)^2 =$ or $(r)^2 =$

In circle M and circle C, the location of the point on the circle was in a different location, but the equations that can be used to determine the length of the radius of the circles are the same.

4. The center of circle R is located at $(3,1)$. Point V lies on circle R at the point $(x, -y)$. Right triangle RZV is drawn inside the circle, such that $\overline{RV}$ is a circle's radius and point Z, $(x, 1)$, lies on radius $\overline{RT}$.



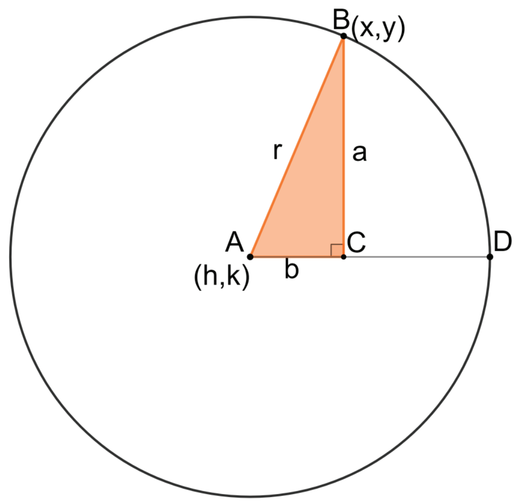
a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{RZ}$	
$\overline{ZV}$	

b. Complete the equation that can be used to determine the length of the radius of circle R.

$(r)^2 = (\qquad)^2 + (\qquad)^2$

5. The center of circle A is located at (h, k). Point B lies on circle C at the point (x, y). Right triangle ABC is drawn inside the circle such that $\overline{AB}$ is a circle's radius and point C, lies on radius $\overline{AD}$. $AC = b$, $BC = a$, and $AB = r$.



a. Use (x, y) and (h, k) to determine the ordered pair for point C.

b. Fill out the table by using the ordered pairs to complete each equation.

Line Segment	Distance Between Endpoints
$\overline{AC}$	$b =$
$\overline{BC}$	$a =$

The Pythagorean Theorem can be used to write the equation, $a^2 + b^2 = r^2$.

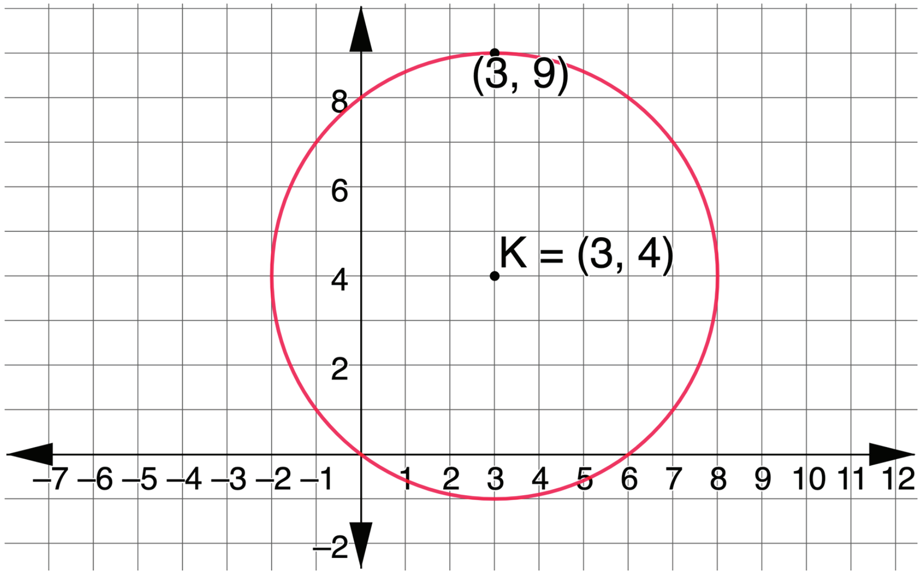
c. Use the equations in the table to substitute for a and b in the equation $a^2 + b^2 = r^2$.

The **center-radius form** of an equation of a circle with center, (h, k), and radius, r, is $(x - h)^2 + (y - k)^2 = r^2$.

12.7.3 Guided Instruction: Key Features of a Circle

Circles have two key features: the center and the radius. The key features can be used to write the equation, $(x-h)^2+(y-k)^2=r^2$, of the circle.

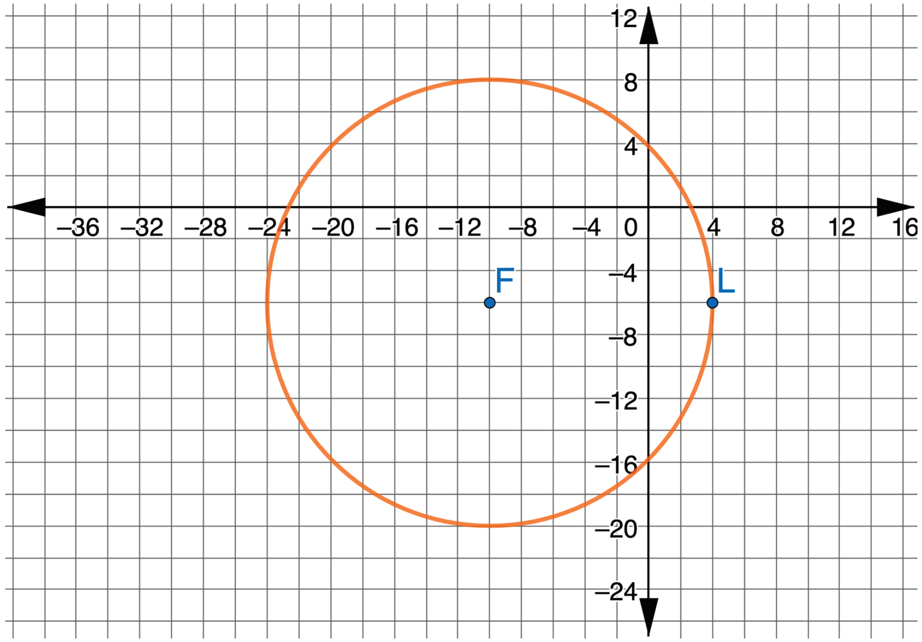
1. Circle K is shown with center $(3,4)$. The point $(3,9)$ is on circle K.



a. Determine the length of circle K's radius.

b. Write the equation that represents circle K.

2. Circle F is shown, where point L is on the circle.



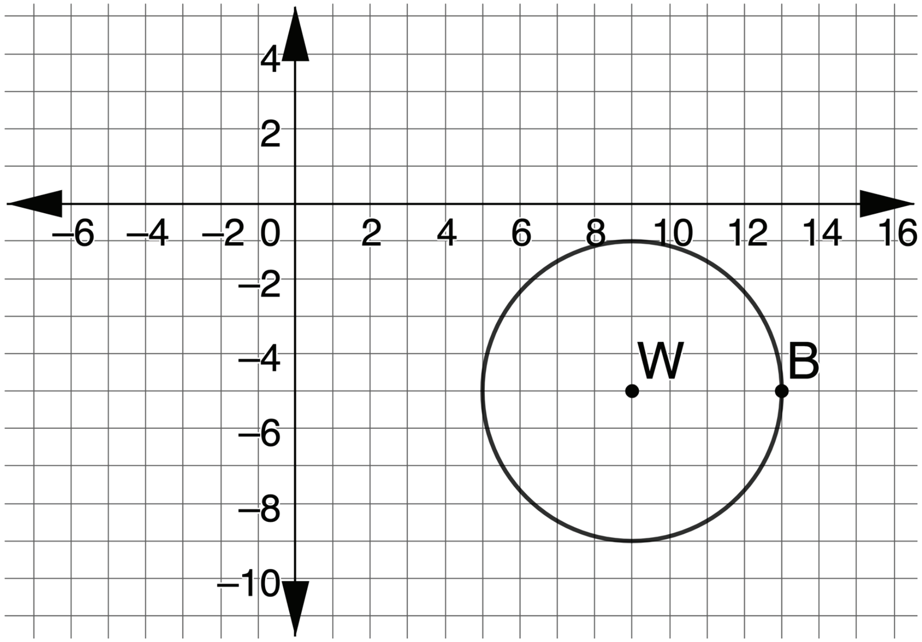
a. Determine the key features of circle F.

Key Features	
Radius	
Center	

b. Write the equation that represents circle F.

12.7.4 Wrap-Up: Ticket Out the Door

1. Circle W is shown, where point B is on the circle.



a. Determine the key features of circle W.

Key Features	
Radius	
Center	

b. Write the equation that represents circle W.

Unit 12, Lesson 8: Standard Form of a Circle

Benchmarks

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

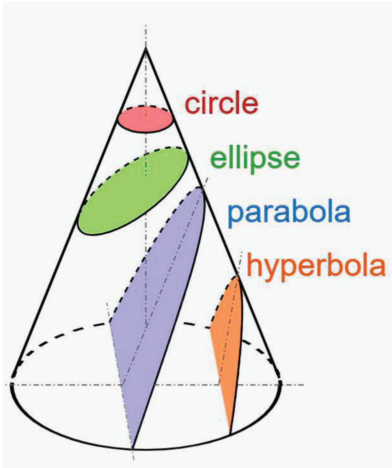
MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.

Learning Targets

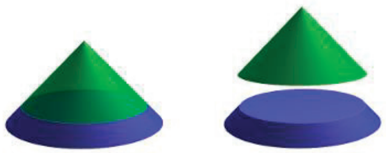
- ☐ I can rewrite the equation of a circle from standard form to center-radius form.
- ☐ I can determine key features of a circle given the equation in standard form.
- ☐ I can write the domain and range using inequality notation, interval notation, or set-builder notation.

12.8.1 Warm-Up: Conic Sections

Conic sections are the family of curves created when a plane slices through a cone. A circle is best known as a conic section. A diagram of five conic sections is shown.



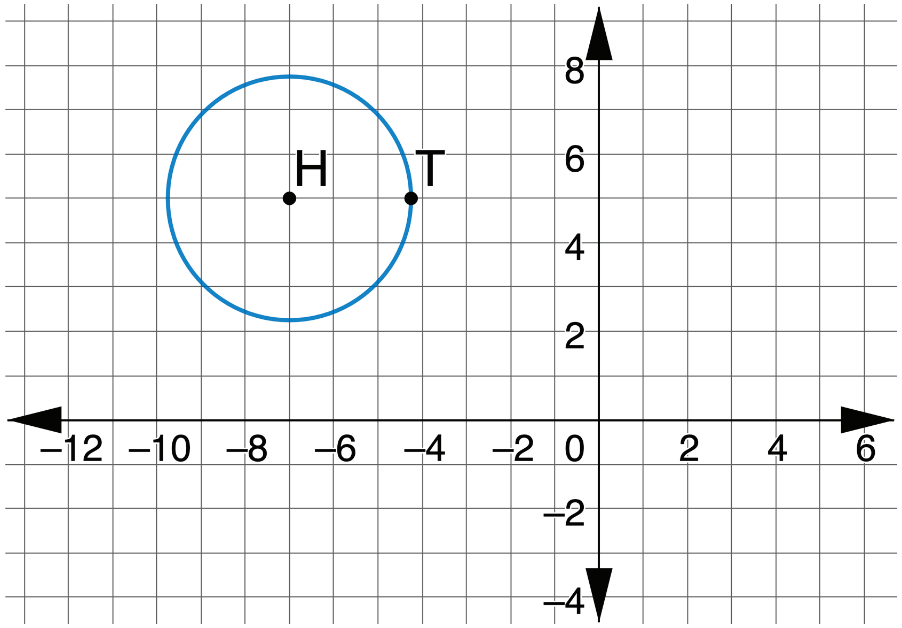
According to Greek philosopher Proclus, Menaechmus, a member of Plato's Academy, discovered conic sections around 350 B.C. Approximately 100 years later, Apollonius, a Greek geometer and astronomer, developed and published the theory of conic sections. His work earned him the title of the Great Geometer.



1. Use the diagram to explain why when a cone is sliced horizontally the result is a circle.

12.8.2 Collaboration: Standard Form

1. The center of circle H is located at $(-7,5)$. Point T lies on circle H at the point $(-4.25, 5)$.



a. Write the equation of the circle in center-radius form, $(x-h)^2 + (y-k)^2 = r^2$.

b. Expand each term of the center-radius form to write an equivalent expression.

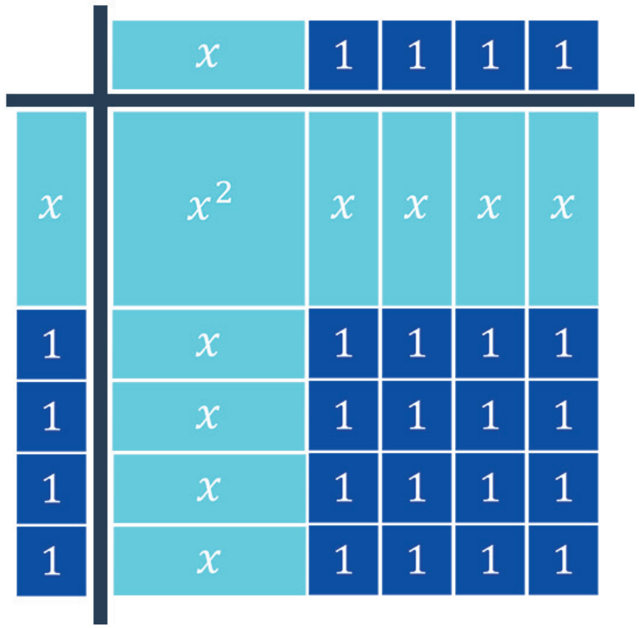
Term	Equivalent Expression
$(x+7)^2$	
$(y-5)^2$	
$(2.75)^2$	

The **standard form** of an equation of a circle is $ax^2 + by^2 + cx + dy + e = 0$.

c. Use the equivalent expressions of the center-radius form to rewrite the center-radius form of circle H in standard form.

12.8.3 Refresher: Completing the Square

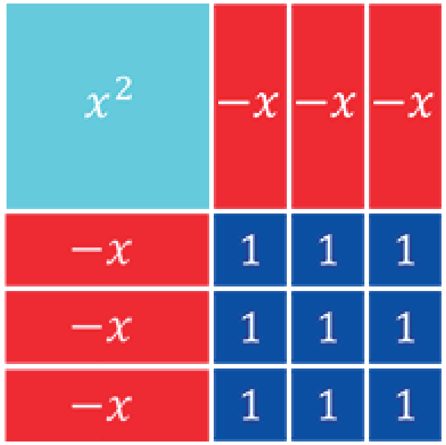
The binomial $(x+4)^2$ can be rewritten as $x^2 + 8x + 16$, which is a perfect square trinomial. The algebra tiles represent both $(x+4)^2$ and $x^2 + 8x + 16$.



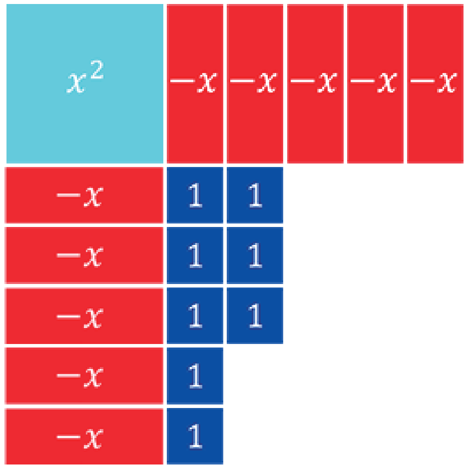
1. Discuss with your partner how the diagram shows that $x^2 + 8x + 16$ is a perfect square trinomial. Summarize your discussion.

2. Discuss with your partner how the diagram shows that $x^2 + 8x + 16 = (x+4)^2$.

3. Write the perfect square trinomial and the binomial squared that are represented by the diagram.



4. The trinomial $x^2 - 10x + 8$ is shown.



a. How many 1 tiles have to be added to complete the square?

Completing the square by adding 1 tiles will change the value of the original trinomial unless the same number of -1 tiles are also added.

b. Complete the table.

Completing the Square	
Original Expression	$x^2 - 10x + 8$
Completing the square	$x^2 - 10x + 8 +$ _____ number of 1 tiles added
Adding the opposite so the expression is equivalent to the original expression	$x^2 - 10x + 8 +$ _____ $+$ _____ number of -1 tiles added
Rewriting so that there is a perfect square trinomial	$(x^2 - 10x + 25) - 17$ perfect square trinomial
Writing the perfect square trinomial as a binomial squared	$(x-5)^2 - 17$

12.8.4 Guided Instruction: Rewriting the Standard Form of the Equation of a Circle

There are several ways to mathematically represent the domain and range.

Written Description	Inequalities	Set-Builder Notation	Interval Notation
All real numbers greater than or equal to -5 and less than or equal to 5	$-5 \leq x \leq 5$	$\{x: -5 \leq x \leq 5\}$, $\{x \mid -5 \leq x \leq 5\}$, $\{x: x \geq -5 \text{ and } x \leq 5\}$, or $\{x \mid x \geq -5 \text{ and } x \leq 5\}$	$[-5,5]$

Rewrite each standard form equation of a circle to center-radius form of a circle. Then, give the key features of the circle. Write the domain and range using interval notation and set-builder notation.

1. $x^2 + y^2 - 14x + 6y + 26 = 0$

<table>
<tr><th rowspan="1">Center-Radius Form</th><th>Center</th><th>Radius</th></tr>
<tr><td rowspan="3"></td><td></td><td></td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td></td><td></td></tr>
</table>

2. $x^2 + y^2 - 18x - 10y = -52$

<table>
<tr><th rowspan="1">Center-Radius Form</th><th>Center</th><th>Diameter</th></tr>
<tr><td rowspan="3"></td><td></td><td></td></tr>
<tr><th>Domain</th><th>Range</th></tr>
<tr><td></td><td></td></tr>
</table>

12.8.5 Guided Practice: Converting Forms of a Circle

1. Consider the equation of the circle represented by $(x+8)^2+(y+11)^2=14^2$.

   a. Rewrite $(x+8)^2+(y+11)^2=14^2$ in standard form.

   b. Give the key features of $(x+8)^2+(y+11)^2=14^2$

Center	Diameter	Radius
Domain	**Range**	

2. Consider the equation of the circle represented by $x^2+y^2-20x=-36$.

   a. Rewrite $x^2+y^2-20x=-36$ in center-radius form.

   b. Give the key features of $x^2+y^2-20x=-36$

Center	Diameter	Radius
Domain	**Range**	

12.8.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.

I need help. I got it. I could help someone.

Rewriting center-radius form to standard form	
Key features of a circle	
Completing the square	
Rewriting standard form to center-radius form	
Set-builder notation	
Interval notation	

Unit 12, Lesson 9: Graphing Circles

Benchmark

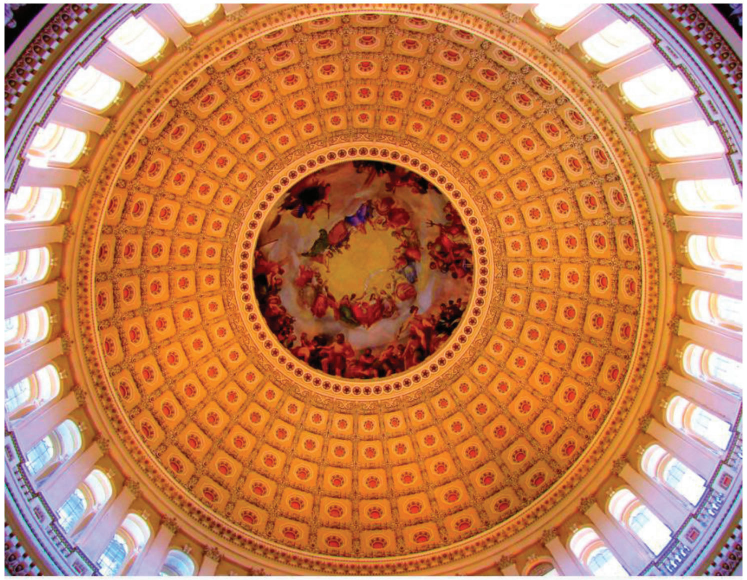
MA.912.GR.7.3 Graph and solve mathematical and real-world problems that are modeled with an equation of a circle. Determine and interpret key features in terms of context.

Learning Targets

- ☐ I can graph a circle and determine its key features when given an equation of a circle.
- ☐ I can solve mathematical and real-world problems that are modeled with an equation of a circle.

12.9.1 Warm-Up: Notice and Wonder

The dome of the U.S. Capitol Building is 288 feet (ft) in height and 96 ft in diameter. The dome covers the Rotunda, a room with a circular plan. The architect created an orchestration of architecture, sculpture, and painting in the U.S. Capitol Rotunda.



1. What do you notice?

2. What do you wonder?

12.9.2 Guided Practice: Graphing Circles

1. The equation of circle W is shown.

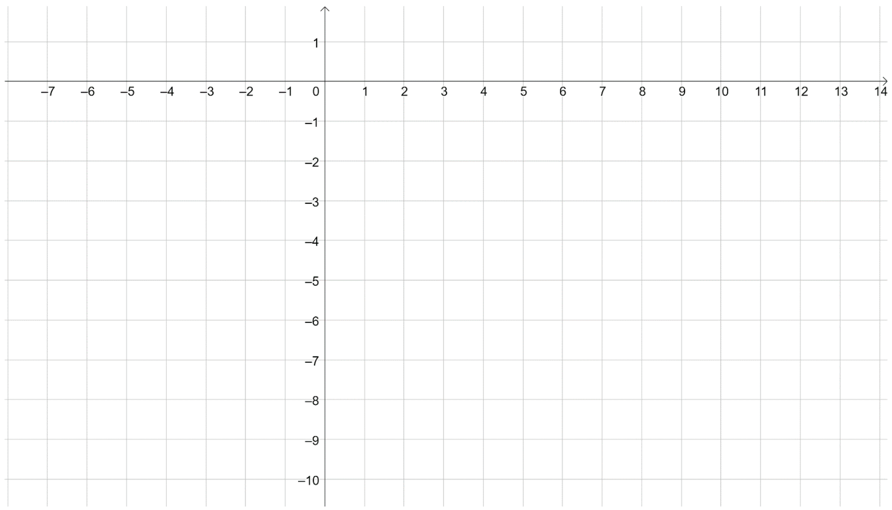
$$x^2 + y^2 - 4x + 10y + 13 = 0$$

   a. Write the equation in center-radius form.

   b. Complete the table.

Center	Radius

   c. Graph the circle.

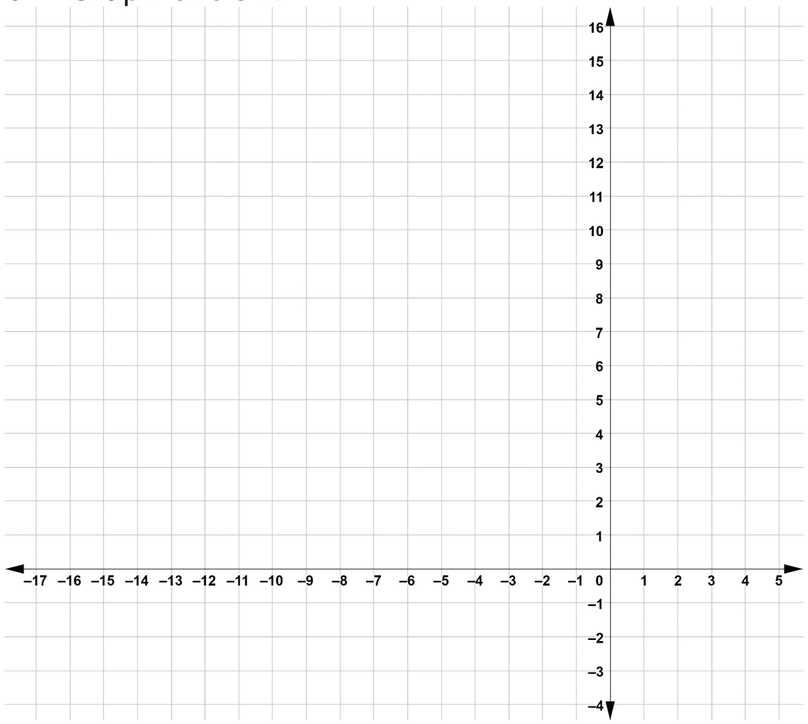


d. Find the domain of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

e. Find the range of $x^2 + y^2 - 4x + 10y + 13 = 0$. Write it as an inequality.

2. Marcus wanted to order from a food delivery service and used the equation $(x+4)^2 + (y-7)^2 = 64$ of circle M to find food within his radius, in miles.

a. Graph circle M.



b. What does the center represent in the context?

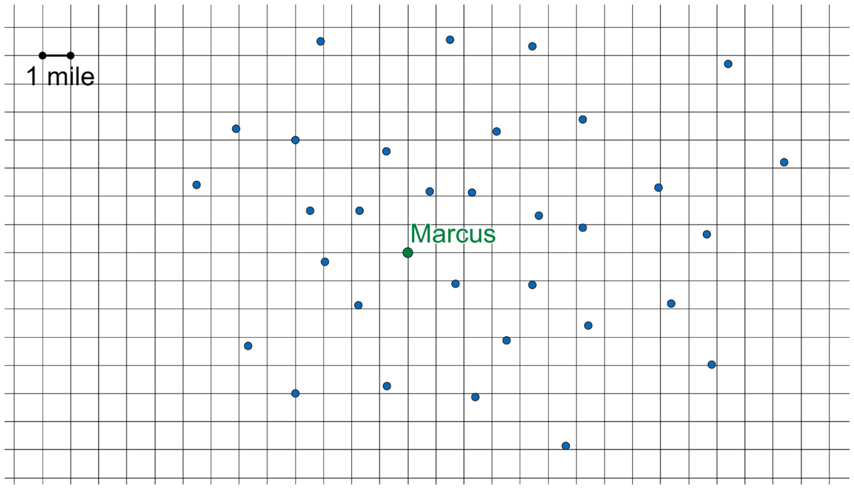
c. What does the radius represent in the context?

d. If the service charges $\$3.25$ for delivery for every four miles driven, what is the most that Marcus could be charged for the delivery?

e. Determine the domain of $(x+4)^2 + (y-7)^2 = 64$. Write the domain in set-builder notation.

f. Determine the range of $(x+4)^2 + (y-7)^2 = 64$. Write the range in set-builder notation.

g. The grid shows Marcus' location in green. The blue dots are the locations of restaurants that deliver using UberEats. Use the grid to demonstrate how Marcus could use the domain and range to determine which restaurants he can order from.



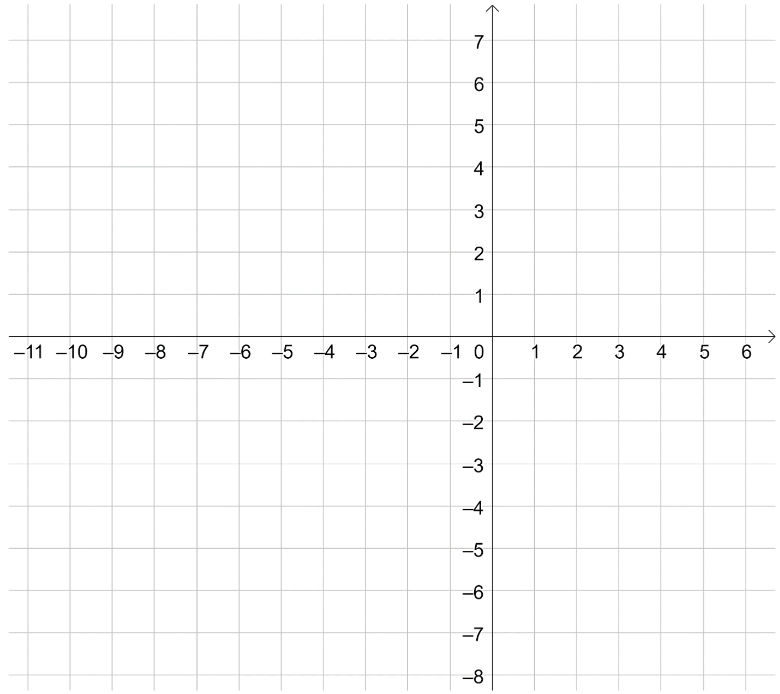
12.9.3 Your Turn!

1. The equation of circle R is $(x+2)^2 + y^2 = 36$.

a. Complete the table.

Center	
Radius	
Domain	____ $\leq x \leq$ ____
Range	____ $\leq y \leq$ ____

b. Graph circle R.

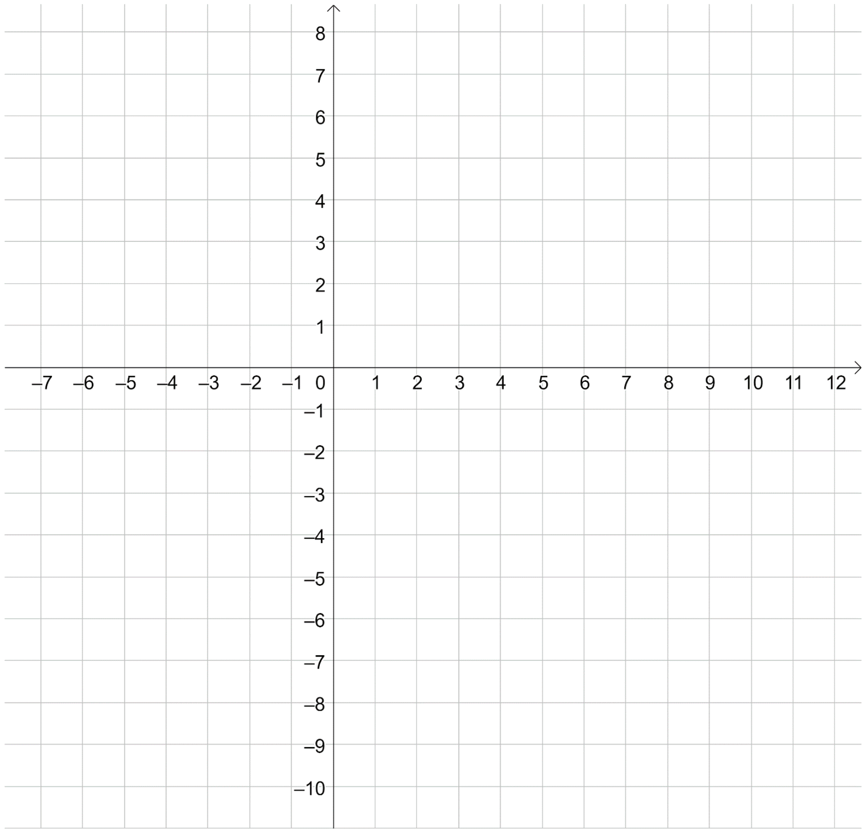


2. Alyona used the equation $(x-3)^2+(y+1)^2=81$ to determine how far she could sit, in meters, from her Xbox and still use her wireless controller.

a. Complete the table.

Center	
Radius	
Domain	____ $\leq x \leq$ ____
Range	____ $\leq y \leq$ ____

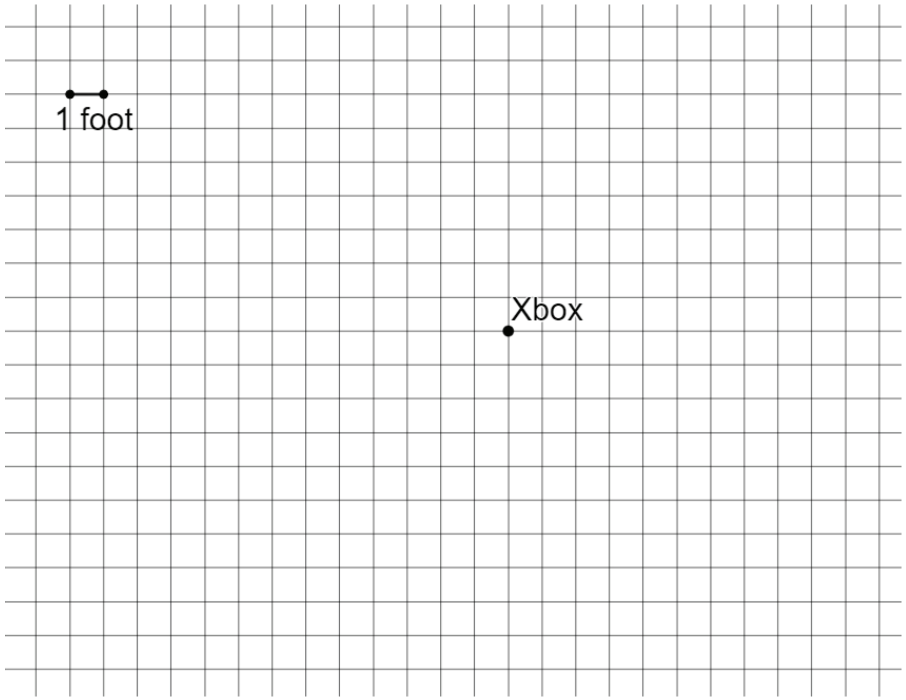
b. Graph $(x-3)^2+(y+1)^2=81$



c. What does the center represent in the context?

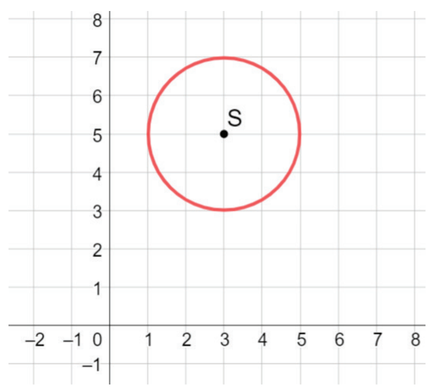
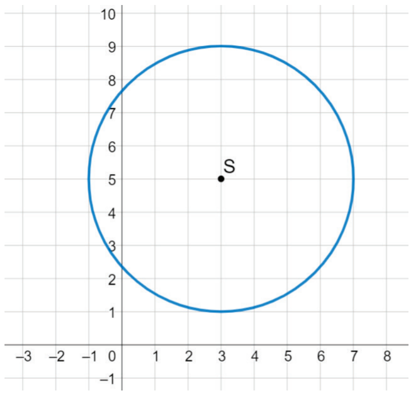
d. What does the radius represent in the context?

e. The grid shows the location of Alyona's Xbox. Use the domain and range to show all of the locations at which Alyona can sit.



12.9.4 Wrap-Up: Error Analysis

1. Ziquan and Amber both graphed circle S, $(x-3)^2 + (y-5)^2 = 4$, but their circles looked different.

Ziquan's Circle	Amber's Circle
	

   a. Who correctly graphed circle S?

   b. Write a note to the person who graphed circle S incorrectly explaining their mistake.

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